

Underbalance or Extreme Overbalance

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Summary

Application of an extreme overbalance, sufficient to fracture the reservoir formation, applied either during or after perforating was first presented as an alternate completion technique during the 1993 Fall SPE Conference.¹ During the ensuing two years, a number of jobs have been run by different operators in a variety of formations with varying applications. The primary question asked by an operator relates to the benefit of extreme overbalance (EOB) versus current perforation techniques of underbalance (UB) or modest overbalance for well control. Of course this question spawns a whole set of new questions about when and how to use and the benefits and risks of the EOB. EOB, as used in this paper, includes extreme overbalance perforating and rapid overpressured perforation extension. This paper compares and contrasts UB and EOB strengths and weaknesses and provides recommendations of when and why to use the different techniques. Field examples are provided of successful and unsuccessful application of EOB completions.

Introduction

Perforating, fracturing and acidizing are simply a means of providing and enhancing flow communication between the reservoir and the wellbore. The objective of these different completion operations is to cost effectively maximize the producibility of the well. Perforating is used to provide the basic fluid communication between the reservoir and the wellbore. Underbalance perforating and acidizing are typically used to bypass or eliminate near wellbore damage. Fracturing is generally used to expose a larger reservoir flow area to the wellbore. Use of propellant tools and now extreme overbalance (EOB) produce short fractures and thus complements underbalance (UB) perforating and acidizing to bypass wellbore damage. However, the process of perforating also creates a permeability damaged region around the perforation commonly called the "crushed zone." Underbalance perforating has been used to remove the perforating damage zone in reservoirs with sufficient pressure and permeability.

Since perforating is the primary means to provide the initial fluid communication with the reservoir and where applicable is performed underbalance, the relevant issue is when-where-how to use EOB as an addition to perforating.

We will first review where UB perforating is applicable, then discuss the interaction between the perforation and the fractures from an EOB process and the implications/questions with regard to the fracture flow path. Several field examples will then be discussed. Finally, based on the physics of EOB, recommendations of applicability will be made. One word of caution: the following discussion on underbalance perforating does not apply to unconsolidated or very weak rock because of the potential failure of the rock from a high underbalance.

Underbalance Perforating

Perforation Damage. The primary purpose of underbalance perforating is to minimize perforation damage. However, there are two different types of perforation damage. The primary one is a damage region that surrounds the perforation tunnel having reduced permeability. This damaged region has historically been called the crushed zone. This damage is a result of fractured sand grains reducing the size of the pore throats. The other, and normally ignored, perforation damage is caused by residual comminuted sand not removed from the perforation tunnel. This material, except at the very end of the perforation tunnel, typically has high permeability and porosities and does not impede production. However, it can restrict injectivity and require higher injection pressures. Removal of this residual comminuted sand is a function of underbalance, charge type and rock strength. Low underbalances, big hole charges, weak rock and downside oriented perforations enhance this residual comminuted sand damage.

Removal of Crushed Zone Damage. Determination of sufficient underbalance to remove the perforation crushed zone permeability damage is still subject to a number of uncertainties. King, Anderson, and Bingham² published field data of underbalance versus permeability with acid improvement/no improvement as a criteria for sufficient underbalance without the need for acidization. Using laboratory data, Behrmann³ published equations for optimum underbalance versus permeability, porosity and perforation hole size to obtain nondamaged perforations. These underbalances are typically three to five times greater than the King data which suggests that even acidization of sandstones does not remove all the perforation damage. Curve fit of permeability-porosity data from Alaska and North Sea Brent reservoirs,⁴ Fig. 1, applied to the equations in Ref. 3, provides a simple underbalance-permeability relationship, Fig. 2 and Eq. (1),

$$p = d^{0.3} [90.4 - 7.86 \ln(k)]^2, \quad (1)$$

where d = perforation tunnel diameter in the rock, inches; k = reservoir permeability, md; and p = optimum underbalance, psi.

The perforation tunnel diameter is not recorded on any data sheets and the casing entrance hole (EH) diameter has been used in the past as a default value. However, the hole diameter in the rock is a function of the charge size and the unconfined compressive rock strength (UCS). An empirical relationship has been developed for average tunnel diameter as a function of casing hole size and rock unconfined compressive strength,³ Eq. (2)

$$d = e_n^* [3.27 - 0.61 \ln(S_{uc})] \quad (2)$$

for deep penetrating charges, where e_n is the casing EH in N80 casing, and S_{uc} is the UCS in kpsi.

Applying the same analysis to the quartz arenites, Fig. 1, gives the following underbalance-permeability relationship, Fig. 3 and Eq. (3):

$$p = d^{0.3} [50.6 + 3.44 \ln(k)]^2. \quad (3)$$

Eqs. (1) and (3) should be used with caution since they are extrapolations of laboratory data on outcrop rocks to reservoir rock using permeability-porosity relations with large data scatter. However, limited laboratory data on low permeability reservoir quartz arenites⁵ and a clean low permeability outcrop rock at 3,000 psi underbalance gave zero to negative perforation damage skins,

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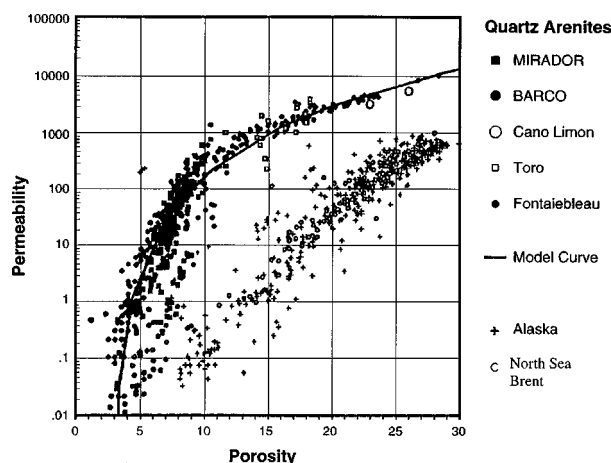


Fig. 1—Permeability-porosity relationships.

Table 1, which suggests that clean (clay free) sands can use lower underbalances as proposed in the above analysis.

Removal of Residual Comminuted Sand. During underbalance operations, varying amounts of the comminuted sand in the perforation tunnel are removed. Although an extensive study of this “injectivity” damage has not been made, limited laboratory experiments have shown that it is more difficult to remove this residual sand under the following conditions:

- Low underbalance,
- use of big hole charges versus deep penetrator charges,
- weak rock,
- downside perforations, and
- single phase oil flow versus two phase oil/brine flow.

Except for very hard rock, the tunnel diameter in the rock is larger than the casing entrance hole. As a result, the hole through the casing/cement provides a flow restriction during the underbalance surge and may prevent all the comminuted sand being swept from the perforation tunnel. Laboratory experiments have shown that the presence of comminuted sand filling at least 50% of the

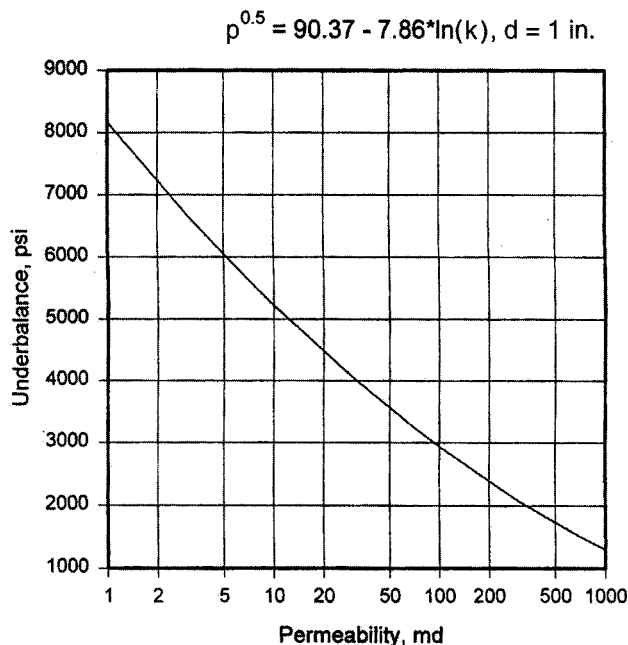


Fig. 2—Optimum underbalance vs. permeability (Alaska and North Sea Brent).

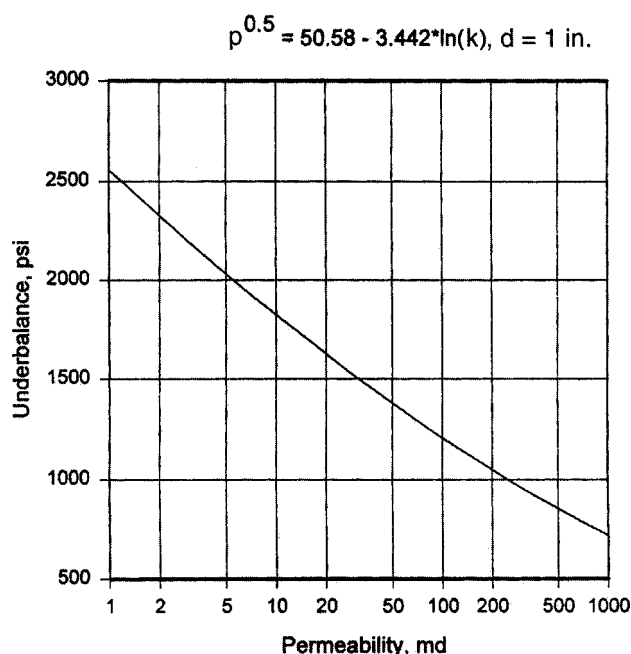


Fig. 3—Optimum underbalance vs. permeability (quartz arenites).

perforation cavity can substantially reduce injectivity without affecting producibility. A specific laboratory experiment is discussed in Appendix A.

When To Perforate Underbalanced. To minimize the crushed zone perforation damage in naturally completed producers and injectors, perforating should be underbalanced when there is sufficient reservoir pressure and no major operational constraints. When less than the optimum underbalance is used, the equations given in Ref. 3 can be used to estimate the residual perforation damage and decide if a stimulation operation is required/justified.

It is also suggested that for all injectors and producers that will be hydraulic or dynamic fracture stimulated, that underbalance perforating be performed prior to a fracture operation if excessive breakdown/injection pressures are deemed to be a potential problem. This recommendation assumes there is sufficient reservoir pressure and permeability to elute most of the comminuted sand from the perforation tunnel. The recommended underbalance given in Eqs. (1) and (3) is generally sufficient to remove most of the comminuted sand debris except for weak rocks.

Extreme Overbalanced Perforating

The Concept. Work by Oryx Energy and Arco has resulted in a new completion technique called extreme overbalance (EOB). Extreme overbalance has been defined as either the application of a very high overbalance pressure during the perforating process (EOP) or very high pressure “surging” of existing perforations. The overbalance, which is significantly above the formation breakdown pressure, creates short fractures in the formation and improves productivity in some wells. EOB has been used successfully in a variety of well environments by a number of operators. More than 800 EOB jobs have been reported, most of these in the U.S. and Canada. While there have been notable successes, EOB is not a global replacement for underbalanced perforating. Rather, it is a complementary process for specific applications.

The EOB approach, developed independently by Oryx and Arco, uses the wellbore tubulars to contain the energy source. The basic technique involves pressuring a large portion of the tubing with gas over a modest amount of fluid. The column of pressurized gas and the liquid beneath it create a high pressure opposite

**TABLE 1—PERFORATION DAMAGE IN CLEAN SANDSTONES
(3,000 psi UNDERBALANCE, 5,500 psi EFFECTIVE STRESS)**

Test	Permeability to Kerosene (md)	Porosity (%)	Charge Weight (g)	Penetration (in.)	Average Hole Diameter (in.)	CFE	Single Shot Skin	Unconfined Compressive Strength (kpsi)
BPC-1	0.96	8.6	23	6.0	0.62	1.67	−0.75	25.1
BPC-3	17.75	8.28	14	5.13	0.62	0.84	0.34	13.58
BPC-5	7.66	12.95	3.5	1.9	0.29	5.82	−2.1	16.06
BPC-5'	9.17	12.95	6.5	3.0	0.36	6.74	−2.0	16.06
BPC-8	4.72	7.9	6.5	2.6	0.25	0.016	170	20.88
CT-1	10.4	14.78	3.5	3.0	0.28	1.08	−0.23	8.8
CT-2	10.4	14.78	3.5	3.15	0.31	1.47	−1.0	8.8

the perforated zone. Field experience, supported by laboratory test data, indicates that the overbalance pressure in the perforation zone should be in the range of 1.4–2.0 psi/ft of well depth.

For wells without existing perforations, the high pressure is applied instantaneously to the formation upon firing the gun, and the wellbore fluid is driven into the perforations. The expanding gas maintains the wellbore pressure in the perforated zone to ensure sufficient loading time to create short fractures (in the order of 10 ft) into the formation. The high downhole flow rates during EOB are not typically attainable in conventional hydraulic fracturing operations. The EOB event lasts tens of seconds.

The basic technique lends itself to a number of interesting variations. Perforating guns may be either tubing or wireline conveyed. Liquids in the well can range from clear brines to acid, to fracturing gels, to gels with suspended proppant, to resin for sand control. Alternatively, the wellbore can be filled entirely with gas.

Previously perforated wells may be treated using either a shear disk or a shearable ball seat or similar device placed at the bottom of the tubing. The shear disk may be run in on wireline and set in a profile. The disk is designed to rupture at a predetermined pressure. With the disk in place the nitrogen pressure is increased until the disk ruptures and the gas pressure drives the fluid into the perforations and the formation. The procedure is similar when using a shearable ball seat. When the packer is set, a ball is dropped from the surface to complete the seal. Nitrogen pressure is increased until shear pins fail, the ball seal is released and the well is surged. The pumpout shear disk is also run on tubing and functions on the same principle. The viability of surging existing perforations is demonstrated by the preference of some operators for separate perforating and surging operations even in the case of initial completions.

One Perspective of EOB. EOB may be viewed as a means to obtain short fractures without the need for large amounts of surface equipment. With multiphased guns, it can also be viewed as a dynamic diversion fluid placement technique. Assuming the EOB concept works, then it might have application for the following situations:

- Fracture past extensive wellbore damage from either new or producing wells,
- fracture past the perforation damage when there is insufficient underbalance,
- provide a matrix acid diversion/placement in place of balls or other diversion techniques,
- prehydraulic fracture treatment to break down the formation, and
- enhance communication with natural fractures.

The first two applications require sufficient fracture conductivity to be successful since the primary flow path is now through the

fractures. Bi-wing or multiple fractures will work as long as the fracture conductivity is not compromised by competing multiple fractures. EOB as a matrix acid displacement technique relies on the acid providing fracture conductivity, therefore multiple fractures are preferred. When used as a prehydraulic fracture treatment, initiation of only a bi-wing fracture is desired to minimize near wellbore competition from multiple fractures. For a naturally fractured reservoir, one wants conductive multiple fractures intersecting the natural fractures. Proper application of EOB to all of the above treatments requires an understanding of the fracture mechanics of EOB.

EOB Fracture Mechanics. The fracture mechanisms involved in EOB are similar to conventional hydraulic fracturing in some respects and yet significantly different in others. Among the factors influencing fracture development in EOB are in-situ stress, well-head pressure, volume of gas, formation permeability, shot density, casing entrance hole diameter, shot phasing, orientation of perforations with respect to the preferred fracture plane (PFP) and the number of fractures created.

Hydraulic fracturing is generally characterized by a single large bi-wing fracture along the formation PFP. However, prior to the extension of this bi-wing fracture, there can be many multiple fracture sites depending on the perforation phasing,⁶ Fig. 4. These multiple fracture initiation sites may contribute to near wellbore tortuosity. On the other hand, multiple fractures from EOB typically occur only where perforations are located and on the PFP if a perforation is not near. EOB also allows a larger angle between the PFP and a perforation before the fracture ignores the perforation and initiates a fracture at the wellbore along the PFP, Fig. 5.

The following discussion of EOB fracture mechanics is confined to vertical or near vertical wells and uses EOB laboratory data, one example of which is given in Appendix A.

Multiple Fracture. The dynamics of EOB enhance the creation of multiple fractures from the phased perforations while tending to limit fracture length and width. The multiple fracture geometry may be detrimental to overall fracture treatment if one wants a single bi-wing fracture and care must be exercised in the perforation phasing and orientation to obtain the most favorable fracture arrangement.

There will always be a bi-wing fracture turning into the PFP. In the case of multiple fractures, it is not clear how far the fractures not in the PFP will propagate nor what the fracture conductivity will be. The extension of fractures originally misaligned with the PFP is dependent on reservoir stresses, stored energy and pressure of the gas in the tubing and competition from the fracture in the PFP. The misaligned fractures are shorter and do not intersect the fracture in the PFP.

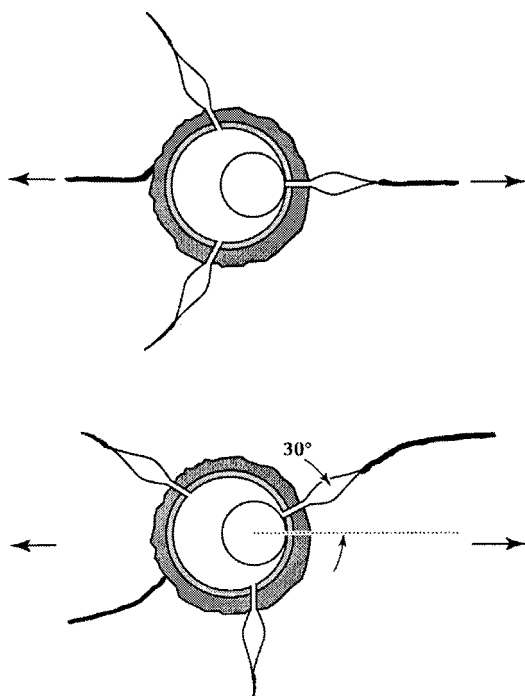


Fig. 4—EOB multiple initiation for 120° shot phasing for two different gun alignments with PFP. Arrows indicate the direction of the PFP.

Fracture Length and Width. The integrated length of all fractures is essentially constant and is not dependent on the number of fractures but rather on the stored energy available before the pressure drops below some minimum value. Fracture width is a function of the individual fracture length and the injection pressure differential above the local formation stress at the perforation. Fracture conductivity is generally considered to be a function of the fracture width, placed proppant, acid or gas erosion of the fracture faces.

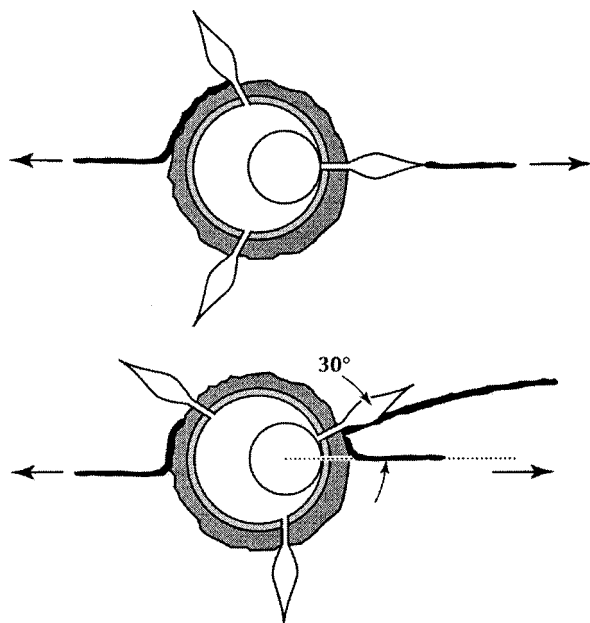


Fig. 5—Hydraulic multiple initiation for 120° shot phasing for two different gun alignments with PFP. Arrows indicate the direction of the PFP.

Fracture Initiation Pressure. Field and laboratory data show that the EOB fracture initiation pressure is always higher than the fracture initiation pressure observed in conventional hydraulic fracturing. EOB also traps residual crushed sand particles in the perforation tunnels (underbalanced perforating tends to flush the particles into the wellbore). The particles may act as a filter cake during injection resulting in a further increase in the dynamic (EOB) fracture initiation pressure.

EOB Field Examples. The following examples refer to Table 2. Each EOB job in the table was chosen to show a wide assortment of well and reservoir conditions and give a broad perspective of where EOB has been attempted and the results of these attempts. In most cases EOB was attempted because conventional completion techniques failed to achieve desired results, or EOB has been successful in a particular surrounding area. The examples and reasons are expressed below.

The Conoco well is considered a success as the production rate achieved from EOB was 50% greater than the production rate of wells in the immediate area. The formation is a very stringy carbonate in which the permeability range is large. The procedure calls for 500 gal of 15% acid to be placed in the bullplugged tubing before the guns are fired.

This procedure takes advantage of the explosively initiated production valve (SXPV) which due to its design opens as the detonation process to the perforating guns is activated. Thus pressure is kept off the casing until the firinghead functions. The operation of the SXPV also ensures that the valve is fully open as the compressed energy in the tubing is released. Therefore, the frictional losses, as the acid is being driven through the tubing and out the SXPV, are held at a minimum. This in turn prolongs the EOB event allowing more energy into the formation which can possibly increase the fracture length. This acid is injected into the reservoir ahead of the N^2 . As the table shows, this is a high permeability carbonate reservoir which yielded very good results.

The Aramco well was a poor performing injector in a friable sandstone. This injector's perforation tunnels were assumed to have sanded up in a short amount of time as the original injection rates dropped substantially in a few weeks, which is a good example of what comminuted sand does to injectors if not properly expelled. Due to poor sand consolidation of the reservoir the needed underbalance pressure to properly expel the sand from the perforation tunnels was not possible without sanding up the string. It was suggested to perforate overbalanced, to stop sand influx into the wellbore, and pump a prepack immediately after perforating. It was believed that the EOB would create propped fractures that would not be as susceptible to sanding due to the larger surface area open to flow. The results of this injector are very good as seen from the table.

The Operator C well is unique due to its high permeability (2,000 md). The EOB job was run with a DST string to be used for fluid loss control and was run in this reservoir as an attempt to improve cleaning of perforation tunnels prior to gravel packing. It was also hoped that the EOB would blow the sand from the perforation tunnels into the formation prior to the trip in and out of the well for gravel pack. This technique had worked well in an offset field with much lower permeability. The operator was warned that the results may not be as successful for this high permeability application, but EOB was decided on to evaluate technique for this field.

The skin is usually quite high in this reservoir due to the partial penetration from the drilling process, and it is difficult to determine what the skin was before the EOB job. An injectivity test afterward was poor at 0.4 bbl/min with 700 psi pump pressure. It was then decided to flow the well with a high drawdown. The

TABLE 2- EXTREME OVERBALANCE JOB DATA

Client	Country	Location	Well Name	Completion Type	Producer/ Injector	Reservoir Type	Depth (ft)	EOB Type	Acid (HCl%)	Permeability (md)
Conoco	USA	North Dakota	Well A	New Well	Producer	Carbonate	9,666	TCP	15	10 to 1,000
Aramco	Saudi Arabia	Saudi Arabia	Hawtah No. 42	Recompletion	Injector	Sandstone	6,500	TCP	...	250
Operator	...	...	Well C	New Well	Producer	Sandstone	9,500	TCP	...	2,000
Petrobras	Brazil	Brazil	Well D	Recompletion	Producer	Sandstone	2,100	Surge	...	20
Petrobras	Brazil	Brazil	Well E	Recompletion	Injector	Sandstone	2,150	Surge	5	20
Operator P	USA	Colorado	Well F	New Well	Producer	Carbonate	1,300	TCP	...	
**Arco	USA	Alaska	Well G	Recompletion	Producer	Sandstone	8,500±	Surge	...	20
Marathon	USA	Texas	Indian Basin No. 9	New Well	Producer	Carbonate	7,837	TCP	20	39

*Offset wells in area typically produce at 2,000 BOPD.

**Pretreatment done prior to hydraulic fracturing.

†In the Petrobras example the existing perforations were deep penetrating charges .39 entry hole with a shot density of 12 SPF at 120° phasing.

‡In the Arco example the existing perforations were deep penetrating charges .45 entry hole with a shot density of 4 SPF at 180° phasing.

injectivity after the drawdown yielded 5 bpm at 800 psi pump pressure. This was considered good. The well was then gravel packed and a production test performed. The production test yielded a low productivity index (PI) with a damage ratio of 7.9. The well was stimulated with a mud acid and the PI was improved by 100%. The damage ratio also was reduced to 3.7. The fact that the well had to be backflowed after the EOB operation convinced the operator that EOB is not applicable in this field.

The examples for the Petrobras wells incorporate the surge technique in which the perforations are already existing across the desired completion interval. The reason EOB was used in this area was due to past problems with fracturing and acidizing. It is believed the EOB technique allows the reduction of skin without attacking primary cementation and the creation of microfractures which do not propagate far enough to allow communication between zones.

A gel-sand cushion is run in the tubing and released through a shear disk at the time the desired nitrogen pressure is achieved. EOB was used on the injector well (well E) because previous completion techniques with acid had broken down the primary cementation allowing undesired communication between zones. As Table 2 shows, the desired injection rate was increased five-fold. The producer, a pumped well, used acid in the tubing at time of isolation valve release and showed a four-fold increase in pro-

duction. These are just two examples of several wells completed with EOB in this area. Results of EOB jobs in this area are considered very good by Petrobras.

The operator F well is a very shallow EOB example where the psi/ft gradient was quite high at 2.1. This well was drilled through a one of a kind formation that showed up on the logs while evaluating a deeper reservoir. The operator in this case has had very good results in microdarcy permeabilities with EOB in an adjacent area and, although the true permeability here was not known, EOB was tried. The reservoir did not respond as well as expected and flowed only 3 BOPD. Acid was then applied as a way to stimulate the reservoir with no increase in production. This is a good case showing where the well was evaluated using a low cost technique instead of using a full Frac job for evaluation which would have been more costly for the operator.

The Arco example⁷ uses their rapid overpressured perforation extension (ROPE) and high energy ROPE (HEROPE) techniques as a pretreatment to hydraulic fracturing. In Arco's case there are several well and reservoir factors they are trying to overcome with EOB. These include deviation, large tubulars, low permeability, small intervals and, due to a water oil contact (WOC), a desire to stay in the oil zone.

TABLE 2- EXTREME OVERBALANCE JOB DATA (continued)

Client	Country	Location	Well Name	psi/ft Gradient	Interval (ft)	Gun Size (in.)	Gun Phasing (degrees)	Shot Density/ Type	Prejob Production	Postjob Production
Conoco	USA	North Dakota	Well A	1.3	16	3 1/2	60	4 SPF/D.P.	*	3,000 BOPD
Aramco	Saudi Arabia	Saudi Arabia	Hawtah No. 42	1.4	72	4 1/2	45/135	12 SPF/B.H.	2,700 BWPD	10,635 BWPD
Operator	...	...	Well C	1.7	40	7	45/135	12 SPF/B.H.	...	?
Petrobras	Brazil	Brazil	Well D	1.4	10	5	120	12 SPF/D.P.	11 BOPD	44 BOPD
Petrobras	Brazil	Brazil	Well E	1.4	10	...	...	...	180 BWPD	1,800 BWPD
Operator P	USA	Colorado	Well F	2.1	90	3 3/8	60	6 SPF/D.P.	...	3 BOPD
**Arco	USA	Alaska	Well G	1.1	20±	...	...	...	0 BOPD	2,300 BOPD
Marathon	USA	Texas	Indian Basin No. 9	1.2	70	3 1/2	60	4 SPF/D.P.	...	5,500 MCFD

*Offset wells in area typically produce at 2,000 BOPD.

**Pretreatment done prior to hydraulic fracturing.

†In the Petrobras example the existing perforations were deep penetrating charges .39 entry hole with a shot density of 12 SPF at 120° phasing.

‡In the Arco example the existing perforations were deep penetrating charges .45 entry hole with a shot density of 4 SPF at 180° phasing.

The ROPE technique has been improved by Arco since its inception to the HEROPE. This technique uses the large tubulars to store large volumes of nitrogen and very little fluid if any in the wellbore at the perforations. Due to the large volume of nitrogen and little fluid in the wellbore, it is believed that the gas flowing through the perforation tunnels and into the formation erodes away any tortuous flow paths, reducing the near wellbore pressure losses (NWPL). These reductions of NWPL have ranged up to 1,350 psi.

The amount of proppant placement seen by Arco since the ROPE inception has gone from just over 50% in the early stages of ROPE to over 95% in 1994. The example well Table 2 well G designed for 34,000 pounds of proppant placed with the actual pounds of prop placement was 39,000 lbm or 115%! Arco's evaluation of the ROPE technique shows that the success rate and the efficiency of the hydraulic fracturing operation and the production rate of the wells are far better than pre-ROPEd wells in the area. The example given in the table is one of the 80 plus wells that have been ROPEd for Arco.

The Marathon example is from a tight carbonate reservoir. The reason EOB was used as a completion technique here was due to the success of another operator in the area and due to lower than expected production rate of the wells in general in this reservoir. Therefore, a new completion technique was considered. The EOB completion technique allowed the well to be surged leaving the well full of nitrogen which could easily be bled off allowing the well to flow. This allowed the well to be put on production much sooner with less cost. Another benefit seen from EOB on this reservoir was a far lower skin than other wells in the area.

Where to Use EOB

EOB Followed by Hydraulic Fracture Stimulation, Modest to Low Permeability. If EOB is to be followed by conventional hydraulic fracture stimulation, the initiation and propagation of multiple fractures should be discouraged. A long bi-wing fracture is needed. Therefore, either a 180° phased gun oriented in the PFP, or a 120° phased gun with random orientation is suggested. Shot density and required casing hole diameter will dictate the charge attributes; generally big-hole charges will work well except where large gun-to-casing clearance exists. Centralization, or partial centralization in the casing is desirable. Shot density will be a function of the fracturing hydraulic horsepower and will generally be modest to low. Direct communication between the perforations and the primary fracture with minimum near wellbore pressure losses is desired. Acid may be used to enlarge the flow path around the casing/cement annulus when the perforations cannot be closely aligned with the PFP. Arco uses a high energy ROPE technique where the amount of liquid is minimized and the amount of pressurized gas is maximized. The assumption is that fracture extension is gas driven and gas erosion of near wellbore tortuous paths eliminates near wellbore pressure losses.

EOB to Overcome Near Wellbore Damage, Modest to High Permeability. Even though the produced fractures are relatively short. EOB treatment can be beneficial in reservoirs where wellbore damage depth is greater than about two thirds of the average perforation depth. High shot density (HSD) guns loaded at high shot phasings (say 45° or 60°) should provide good results taking advantage of multiple drainage paths into the wellbore. The key for this application is for the fractures to have some fluid conductivity. Some operators use proppant filled canisters which open during the EOB process. The theory is that the proppant is entrained in the high velocity flow and erodes the fracture face.

EOB in Carbonate Reservoirs. EOB has been successful in carbonate reservoirs when the wellbore fluid is acid. EOB acts as a dynamic diversion for the acid. HSD guns with many shot phasings are recommended.

EOB in Naturally Fractured Reservoirs. Creation of multiple fractures that intersect the natural fractures provides another EOB application providing there is sufficient fracture conductivity as already discussed. One issue is how far the fractures travel before turning into the PFP, generally the natural fracture orientation. Experiments suggest the fracture starts turning after one wellbore diameter. If the EOB fracture is reoriented in the PFP after two to three wellbore diameters, then the natural fracture spacing should be less than two to three wellbore diameters. High shot density guns with many shot phasings are recommended.

Conclusions

EOB is a completion process that may be used during or after perforating. It is not a replacement for perforating. For new completions (no existing perforations), one can choose between perforating alone, either underbalance or modest overbalance, or EOB during the perforating process. EOB while perforating requires a consideration of completion objectives, reservoir properties, maximum allowable surface pressure, economics, etc., to determine if an integrated approach is better than perforating followed by EOB.

Proper application of EOB, either integrated or segregated from perforating, has been successfully applied in a variety of completions. Laboratory tests and field observations have provided a better understanding of the EOB physics and provided recommendations on choices of perforating systems and application of the process.

Nomenclature

- p = underbalance pressure, psi
- k = permeability, md
- d = tunnel diameter, in.
- S_{uc} = unconfined compressive rock strength, kpsi
- e_h = casing entrance hole in N80 casing, in.

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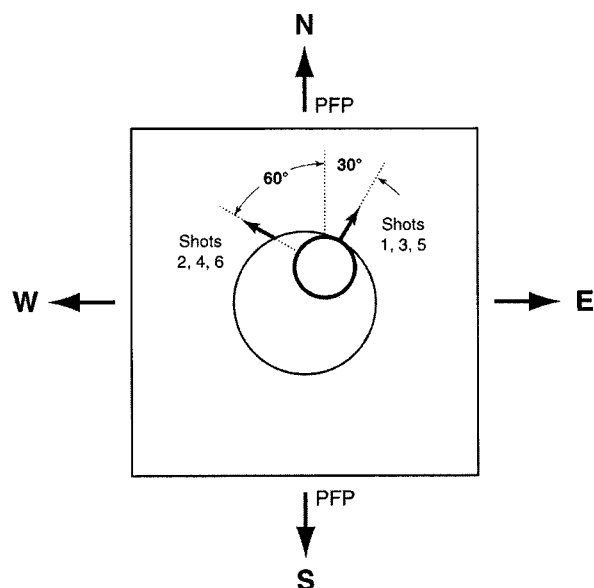


Fig. 6—Gun orientation, Test One.

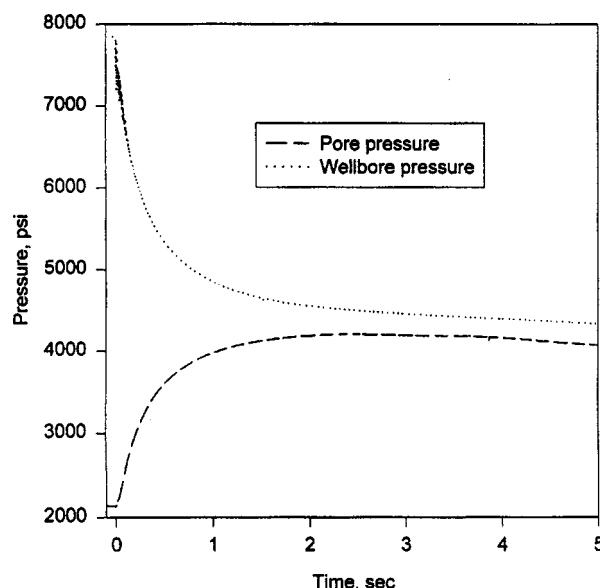


Fig. 7—EOB surge test.

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Appendix A: EOB Laboratory Experiments

Two large Berea block EOB experiments were conducted in the TerraTek polyaxial stress frame to: (1) evaluate the performance of EOB during the perforating event and after perforating underbalanced, (2) evaluate fracture conductivity for very short fractures, and (3) determine if multiple fractures are initiated. Six shot/ft guns with 3.5 g charges with $\pm 45^\circ$ phasing were used. The guns were oriented with one plane of three charges 30° from the preferred fracture plane (PFP), Fig. 6. For both experiments, the horizontal stress differential was 1,000 psi and the differential wellbore-pore pressure was 5,750 psi. Details of the use of this experimental facility can be found in Ref. 6.

Block Preparation. Two in-situ pore pressure probes were mounted in the block, one 3 in. from the wellbore and a second one 3.88 in. from the block outer flow boundary. The block was vacuum brine saturated by flowing 3% KCl from an 18 in. section of the open wellbore to the block boundaries. The block permeability between these two in-situ probes was about 245 md. Although the wellbore was drilled with a diamond coring bit, there was considerable wellbore damage with a permeability of 28 md assuming a damage depth of 1 in. Observation of the split block after dye injection through the perforations shows the damage zone to be about 1 in. giving a wellbore damage zone permeability reduction of about 0.1. The casing was cemented using oilfield cement and allowed to cure with the block under stress.

Test One. Test one was shot with an underbalance of 500 psi, sufficient to elute the comminuted sand from the perforation tunnel but not sufficient to produce a nondamaged perforation. The PI increased from 0.08 to 0.22 cc/s/psi as the total brine flow rate was increased from 3.7 to 60 cc/s. A mini stack (five foot length

of 2.75 in. inside diameter (ID) casing) was placed on top of the existing 2.75 in. ID casing with a break seal separating the two. The top stack was pressurized with argon to 7880 psi (total gas volume at 7880 psi was 1.51 gal). The fluid in the casing across the perforations was 1.51 gal of brine at 2110 psi. The seal was explosively removed and the perforations surged at a differential pressure of 5770 psi. The time history of the wellbore and pore pressures are shown in Fig. 7. The rapid decrease in wellbore pressure is interpreted as fracture initiation. The pore pressure can leaked after the surge and it was not possible to produce the rock under a back pressure to remove the injected Argon gas. The block was removed and mounted in another pore-pressure can and placed back in the stress frame. Brine was flowed for 4 hours with increasing drawdown to 300 psi. Substantial gas was being eluted. The final PI was 0.12 cc/s/psi indicating a two phase liquid/gas

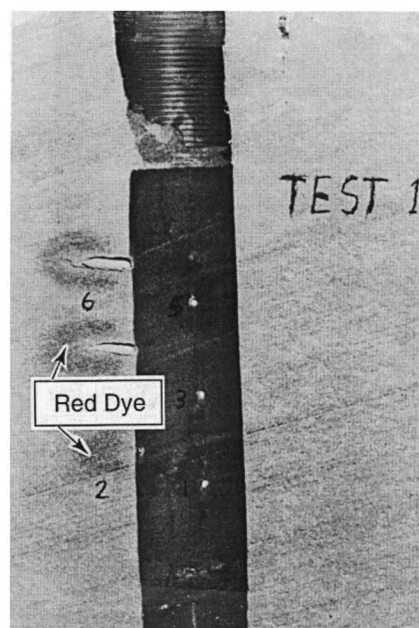


Fig. 8—Photo 1.

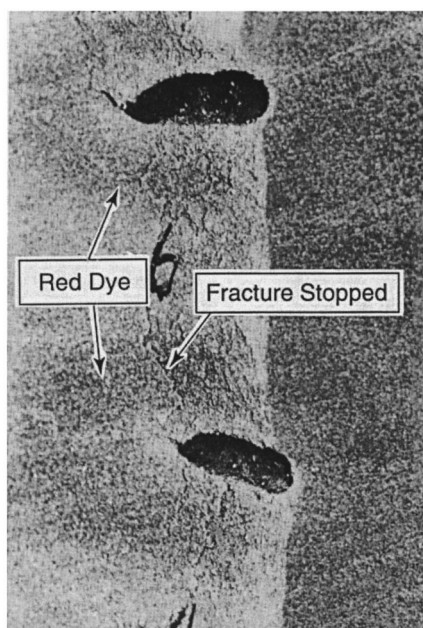


Fig. 9-Photo 2.

relative permeability problem. Red dye was injected at 50 psi for 5 minutes followed by injection of RTV.

Fractures were initiated from perforations 1, 3, and 5 which were 30° from the PFP, between perforations 4 and 6 which were 60° from the PFP and opposite perforations 1, 3, and 5 (called the 180° fracture). **Fig. 8** shows the RTV in the six perforations, the fracture crack between perforations 1, 3, and 5, and the red dye halo around perforations 2, 4, and 6. **Fig. 9** is a close-up of perforations 4 and 6. The fracture extended the length of the perforations, 4 and 6, and stopped. The lack of dye around the first inch of the wellbore suggests wellbore damage. The minimal dye adjacent the perforation suggests the extent of perforation damage. **Fig. 10** is a close-up of the fracture initiation from perforations 1, 3, and 5. **Fig. 11** shows the exposed fracture from perforations 1, 3, and 5. Note that after about one wellbore diameter, the fracture

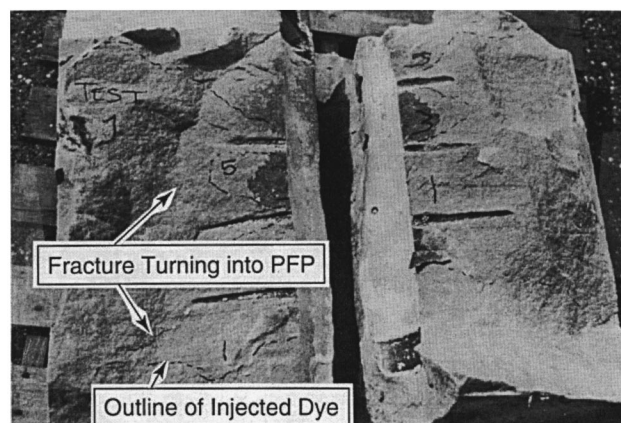


Fig. 11-Photo 4.

TABLE A-1- PRODUCTION-INJECTION FLOW RESULTS, TEST 2

Time (min)	Differential Pressure (psi)	Production Index (cc/s/psi)	Injection Index (cc/s/psi)
0	142	0.255	
41	148	0.243	
0	142		0.165
10	150		0.095
0	148	0.307	
5	146	0.296	
10	299	0.275	
20	300	0.267	
0	143		0.146
6	148		0.091
12	313		0.08
16	280		0.057
20	464		0.053
28	436		0.038
33	589		0.04
41	617		0.03
0	295	0.257	
8	309	0.25	

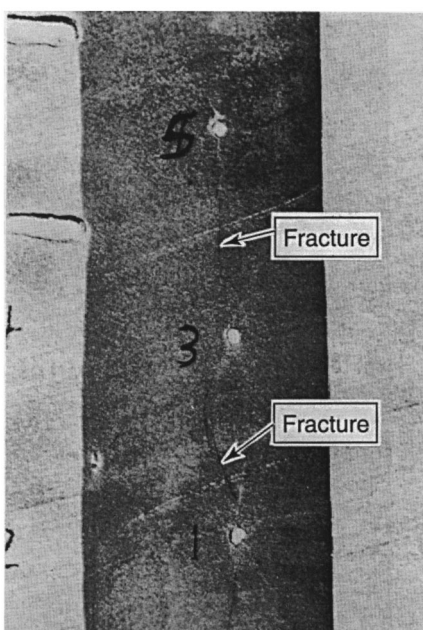


Fig. 10-Photo 3.

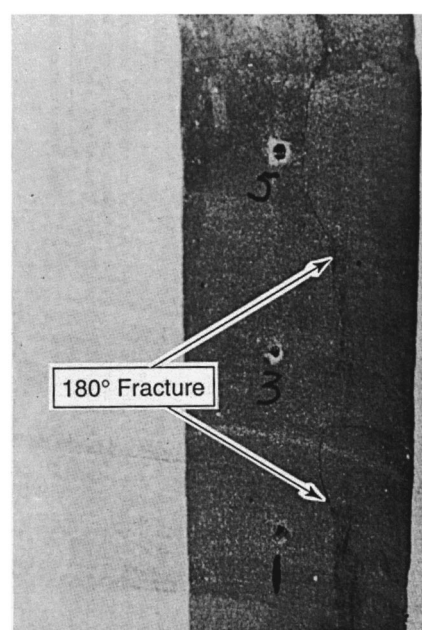


Fig. 12-Photo 5.

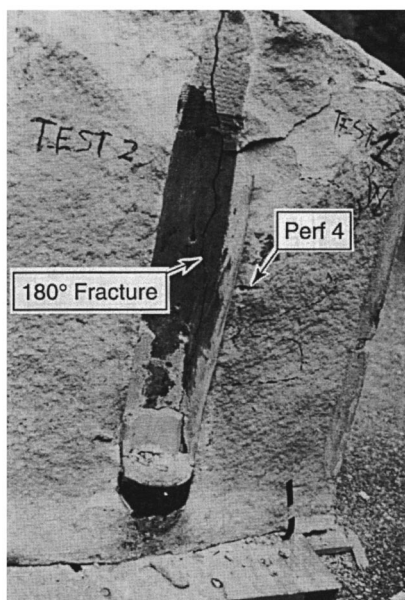


Fig. 13-Photo 6.

turned into the PFP. Fig. 12 shows the 180° fracture initiation. The perforations seen in Fig. 12 are from Test Two. Fig. 13 shows the rock being opened along the 180° fracture.

Test Two. The perforations from Test One were squeezed and sealed from the wellbore. The same gun as used in Test One was also used in this test with perforations 1, 3, and 5 located 30° west of south. The PFP was in the North-South direction. A small stack, 44.5 in. of 9 5/8 in. casing, was placed on the casing. One gallon of red dyed fluid was placed in the stack followed by a gas volume of 11 gal at 7,180 psi. There was no communication with the rock indicating the squeezed perforations were sealed. Pore pressure was increased to 1,440 psi and the gun fired. This was the same differential as for Test One but with seven times more volume of pressurized gas. The rock did not fracture. Fig. 14 shows a modest drop in wellbore pressure and a small increase in pore pressure. Since postshot evaluation showed that perforations were made in the rock, it is clear that the perforations were plugged

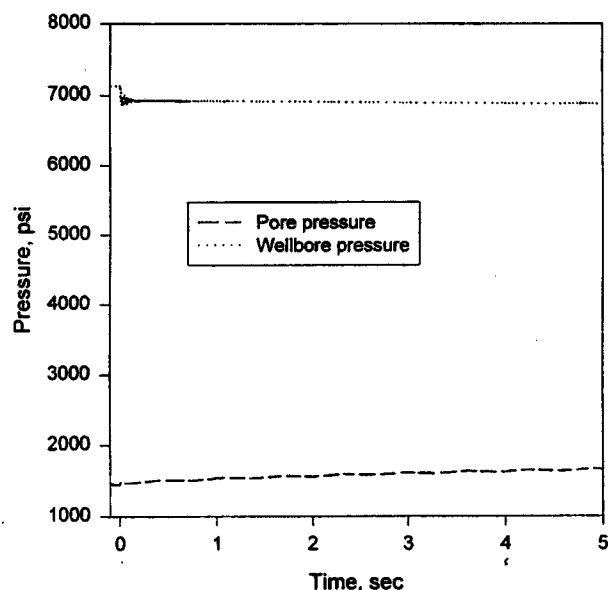


Fig. 14-EOB perforating test.

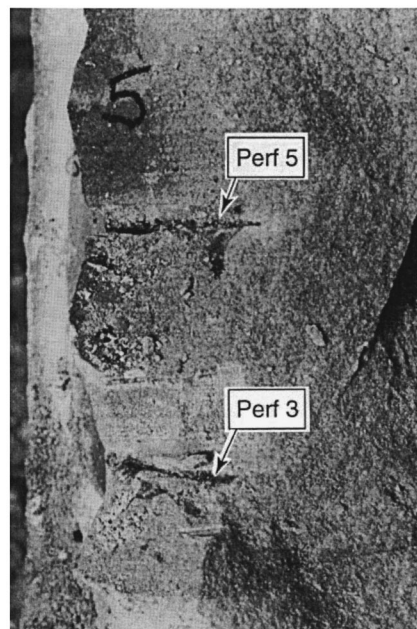


Fig. 15-Photo 7.

with debris and formed an impermeable barrier for injection. The stack was removed and brine produced through the block with a PI of 0.243 cc/s/psi, about 10% greater than for Test One. Although the sand debris formed an external filter cake which prevented injection, it was easily removed upon production. At this point, a series of injection-production tests were performed in hopes of achieving good injection for a subsequent overbalance surge test similar to Test One. Table A-1 summarizes the flow results. Note that injection continued to decline with time and pressure without degrading production. The final injection was only 12% of the production. Fig. 15 shows perforations 3 and 5 from Test Two. The lack of open tunnels is obvious and gives validity to the hypothesis that the comminuted sand debris from the perforation process can form an effective external filter cake resulting in an increased fracture gradient.

Conclusions

These tests demonstrate the following:

- Fractures can be initiated from phased perforations.
- Perforations nearest the PFP will have the dominant fracture, i.e., longest.
- The dominant fracture will create a second fracture opposite it to have a bi-wing fracture.
- Sand debris in the perforation tunnel acts both as an impermeable filter cake and as a pressure barrier resulting in higher fracture initiation pressure.
- Additional multiple fractures, as seen in hydraulic fracture tests, have not been observed.
- A fracture will initiate from a perforation that is at least 30° from the PFP without initiating a fracture from the sand face along the PFP. This is in contrast to hydraulic fractures.
- For these very short fractures, there was zero fracture conductivity. The limited block size restricts the study of fracture conductivity.

SI Metric Conversion Factors

ft × 3.048*
gal × 3.785 412
in. × 2.54*
psi × 6.894 757

E-01 = m
E-03 = m³
E+00 = cm
E+00 = kPa

*Conversion factors are exact

SPEPF

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